

# Python Programming (B.Tech CSE - Sem 2)

## Experiment 03 – Exercise Sheet

### Loops and Basic Programs

Tofik Ali

February 20, 2026

**Repository:** <https://github.com/tali7c/Python-Programming>

**Note:** This document contains only problem statements (no solutions).

## Instructions

- Write a separate Python program for each exercise.
- Use loops and conditionals as needed.
- Validate input where appropriate (example: negative numbers).
- Display output clearly and consistently.

## Exercises

### Exercise 01: Factorial of a Number

Write a Python program to compute the factorial of a non-negative integer  $n$  using a loop.

**Input:** an integer  $n$

**Output:**  $n!$  (and a suitable message if  $n < 0$ )

### Exercise 02: Armstrong Number Check

Write a Python program to check whether an integer  $n$  is an Armstrong number. An Armstrong number is a number equal to the sum of its digits raised to the power of the number of digits.

**Input:** an integer  $n$

**Output:** print whether  $n$  is Armstrong or not

### Exercise 03: Fibonacci Series

Write a Python program to print the first  $n$  terms of the Fibonacci series using iteration.

**Input:** an integer  $n$  (number of terms)

**Output:** the first  $n$  Fibonacci numbers

### Exercise 04: Prime Number Check

Write a Python program to check whether a given integer  $n$  is prime.

**Input:** an integer  $n$

**Output:** print whether  $n$  is prime or not

### Exercise 05: Palindrome Check

Write a Python program to check whether an input string/number is a palindrome (reads the same forward and backward).

**Input:** a string (or number as input text)

**Output:** print whether it is a palindrome or not

### Exercise 06: Sum of Digits

Write a Python program to compute the sum of digits of an integer.

**Input:** an integer  $n$

**Output:** sum of digits of  $n$

### Exercise 07: Numbers Divisible by 5 or 7 (1 to 100)

Write a Python program to generate and display all integers between 1 and 100 that are divisible by 5 or by 7. Also display the total count.

**Input:** none

**Output:** list of numbers and the count

### Exercise 08: Lowercase to Uppercase

Write a Python program to convert an input string to uppercase.

**Input:** a string

**Output:** uppercase version of the string

### Exercise 09: Multiplication Table

Write a Python program to print the multiplication table of a given integer from 1 to 10.

**Input:** an integer  $n$

**Output:** lines of the form `n * i = result` for  $i = 1..10$

### Exercise 10: Pattern Printing

Write a Python program to print the following pattern for  $n = 5$ :

```
123454321
1234 * 321
123  * * 21
12   * * * 1
1    * * * *
```

**Input:** none (or optionally take  $n$  as input)

**Output:** print the pattern line by line

### Exercise 11: Harmonic Series Sum

Write a Python program to compute the sum of the harmonic series:

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

**Input:** an integer  $n$

**Output:** the sum (print up to 4 decimal places)